

International Panel on Fissile Materials

Fostering initiatives to reduce stocks and end the production and use of highly enriched uranium and plutonium

Menu 

IPFM BLOG

Spent fuel reprocessing at Mayak in 2024

AUGUST 21, 2025   

The 2024 annual report of Russia's Federal Nuclear and Radiological Security Program contains data on spent fuel reprocessing at the RT-1 plant at Mayak.

Fuel type	Amount, tonnes
VVER-400	50.9
VVER-1000	46.3
BN-600	20.2
RBMK-1000 (damaged fuel)	16.5
Irradiated standard uranium blocks (OSUB)	0.3
BOR, SM, MIR	0.1
Total	134.3

The "irradiated standard uranium blocks" are fuel elements that were used in plutonium production reactors. BOR, SM, and MIR are research reactors.

The nominal capacity of the RT-1 plant is [400 tonnes of fuel a year](#). Historically, it has been reprocessing about [110-130 tonnes of fuel a year](#).

The document also shows that the cumulative total amount of reprocessed fuel in 2016-2024 was 993 tonnes.

TAGS

BOR reactor, Mayak, MIR reactor, RBMK, Russia, SM-3 reactor, spent fuel reprocessing, VVER

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